🧪Hands-On Lab Exercise 8: File Handling and Text Processing in Shell Script

Topic: Working with Files and Text in Shell Script
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File handling is one of the most important parts of shell scripting because real automation always works with files. Systems store data in files, logs are written in files, configurations are saved in files, and outputs are generated in files. A shell script that cannot handle files is incomplete for real-world automation.

In automation, scripts are not only responsible for running commands, but also for creating files, reading files, updating files, moving files, deleting files, and processing the content inside files. File handling converts a script from a simple command runner into a real automation tool.

Text processing is equally important because most system information, logs, configuration files, and outputs are in text format. Shell scripting provides powerful tools to read and process this text, which allows automation to analyze systems, extract information, and generate meaningful outputs.

In simple words, file handling and text processing allow shell scripts to **work with data**.

📊Learning Objective

• Understand file handling in shell script
• Learn how to create files and directories
• Learn how to write data into files
• Learn how to read data from files
• Learn how to copy, move, and delete files
• Understand basic text processing

🔗Learning Outcome

After this lab, participants will: - Create and manage files - Automate file operations - Read and write file content - Handle system files - Build data-based automation

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📘Basic File Commands in Shell

Some important commands: - `touch` → create file - `mkdir` → create directory - `cat` → read file - `echo` → write text - `cp` → copy file - `mv` → move/rename file - `rm` → delete file - `head` → read first lines - `tail` → read last lines - `wc` → count lines/words

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🧪Example 1: File Creation and Writing Data

Step 1: Create Script

```
touch file1.sh
```

Step 2: Open File

```
nano file1.sh
```

Step 3: Write Script

```
#!/bin/bash

mkdir data
touch data/info.txt

echo "This is my first file" > data/info.txt
echo "Learning file handling in shell" >> data/info.txt
echo "Automation starts with files" >> data/info.txt

cat data/info.txt
```

Step 4: Permission

```
chmod +x file1.sh
```

Step 5: Run

```
./file1.sh
```

🧪 Example 2: Reading Data from a File

```
#!/bin/bash

echo "Reading file content:"
cat data/info.txt


echo "First 2 lines:"
head -n 2 data/info.txt


echo "Last 2 lines:"
tail -n 2 data/info.txt
```

🧪 Example 3: Copy, Move, and Delete Files

```
#!/bin/bash

cp data/info.txt data/info_copy.txt

echo "File copied"

mv data/info_copy.txt data/final.txt

echo "File renamed"

rm data/final.txt

echo "File deleted"
```

🧪 Example 4: Basic Text Processing

```
#!/bin/bash

file=data/info.txt

echo "Line count:"
wc -l $file

echo "Word count:"
wc -w $file

echo "Character count:"
wc -c $file
```

🧪 Example 5: Searching Text in File

```
#!/bin/bash

grep "shell" data/info.txt
```

Understanding the Flow

Scripts now interact with data, not just commands. They create files, store information, read it, process it, and manage it. This is the foundation of real automation systems.

Key Understanding

File handling transforms scripts into data-driven systems. Text processing transforms scripts into intelligent automation tools.

📌 Summary

- Files are core to automation
- Data handling is essential

- Scripts must manage files
- Text processing enables intelligence
- Real automation is data-driven

💴End of Lab

This lab builds the **data foundation of automation**.

Now scripts can: - Handle data - Process information - Manage logs - Manage configurations - Build real systems

This is essential for: - DevOps automation - Monitoring systems - Log analysis - Backup automation - Infrastructure management

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